

Algebra II Logs and Series Practice Test

Instructions

Complete in 60 minutes. Show clear work. Use exact values unless a decimal approximation is explicitly requested. For logarithmic equations, state domain restrictions and check for extraneous solutions.

Reference Facts

- Log properties:

$$\log_b(MN) = \log_b M + \log_b N, \quad \log_b\left(\frac{M}{N}\right) = \log_b M - \log_b N, \quad \log_b(M^r) = r \log_b M$$

$$\log_b(b^x) = x, \quad b^{\log_b x} = x, \quad \log_b x = \frac{\log_a x}{\log_a b}$$

- Arithmetic and geometric sequences:

$$a_n = a_1 + (n - 1)d, \quad S_n = \frac{n(a_1 + a_n)}{2}$$

$$g_n = g_1 r^{n-1}, \quad S_n = \frac{g_1(1 - r^n)}{1 - r}, \quad S_\infty = \frac{g_1}{1 - r} \quad (|r| < 1)$$

- Consecutive integer sum:

$$1 + 2 + 3 + \cdots + n = \frac{n(n + 1)}{2}$$

Problems

1. Determine whether the relation is a function. Then determine whether its inverse relation is a function. Explain briefly.

$$(-4, 9), (-2, 3), (0, -1), (2, 3), (4, 9)$$

2. Let

$$f(x) = \frac{x+6}{x-2}.$$

Find the domain, range, x -intercept, y -intercept, and both asymptotes.

3. Determine whether each function is even, odd, or neither. Justify algebraically.

$$f(x) = x^5 - 4x^3 + x, \quad g(x) = x^4 - 3x^2 + 8, \quad h(x) = x^3 + x^2 - 2x.$$

4. Expand completely:

$$\log_b \left(\frac{16x^3 \sqrt{y}}{b^4 \sqrt[3]{z^5}} \right).$$

5. Condense into a single logarithm:

$$2 \log_5(x-1) - \frac{1}{2} \log_5(y+3) + 3 \log_5(25) - \log_5(z^2).$$

6. Solve and check for extraneous solutions:

$$\log_3(x-2) + \log_3(x-6) = 2.$$

7. Solve and check for extraneous solutions:

$$\log_2(x+4) - \log_2(x-1) = \log_2(x-2).$$

8. Evaluate exactly.

$$\sum_{k=1}^{30} (4k-7), \quad \sum_{j=8}^{42} j$$

9. For the arithmetic sequence with $a_5 = 19$ and $a_{18} = 71$, find a_1 , d , and S_{25} .

10. For the geometric sequence with $g_3 = 48$ and $g_7 = 3$, find all possible values of r and g_1 . Then find g_{10} for each possible sequence.

11. Evaluate exactly.

$$\sum_{n=0}^7 6 \left(\frac{2}{3} \right)^n, \quad \sum_{k=1}^n (5k-2)$$

12. Determine whether each infinite geometric series converges. If it converges, find its exact sum.

$$\sum_{n=1}^{\infty} 18 \left(-\frac{1}{3} \right)^{n-1}, \quad \sum_{n=0}^{\infty} 5 \left(\frac{4}{3} \right)^n$$